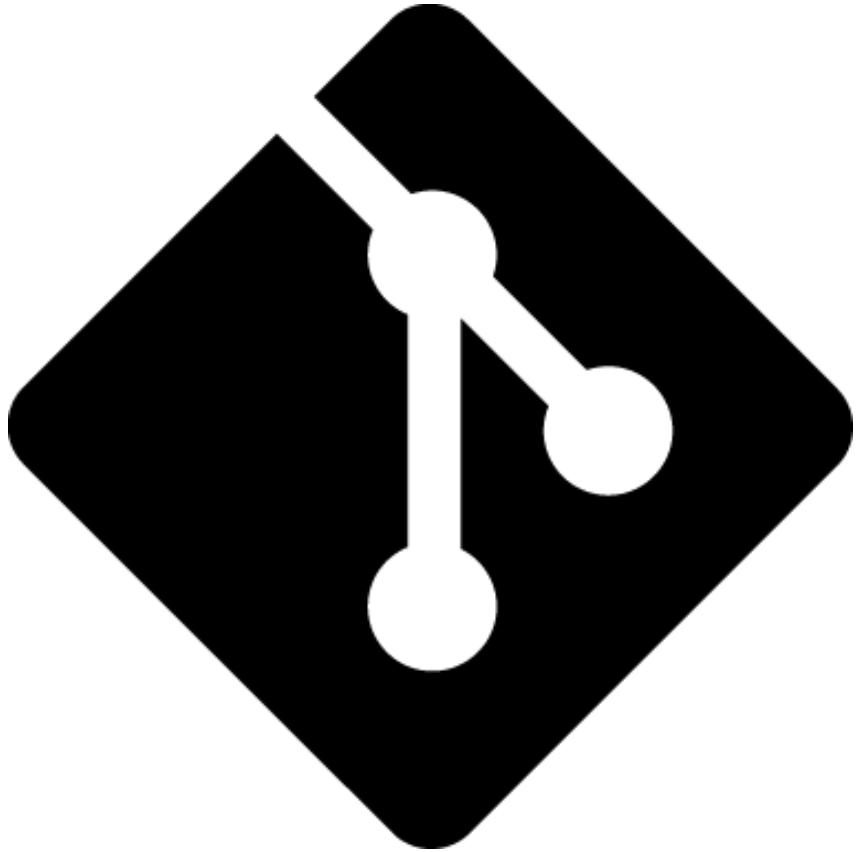

Git Introduction Course

Romain De Groote

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- Track changes
- Revert to previous versions
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- Use a cloud storage service?
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Real difficulty: organization between the members to know who works on what, and to avoid conflicts when merging changes.

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Online systems are not protected from all vulnerabilities, so try to always have a local copy of your most important works.

GitHub status

GitLab status

Cloning a repository

Practise exercise:

- Open Git Bash
- Use `cd <path>` to navigate to the folder where you want to clone the repository
  - try `ls` to list the content of the folder and `cd ..` to go back to the parent folder
- Use `git clone https://github.com/MatsuneMikuroi/GitCrashCourse.git <folder_name>` to clone the repository into a specific folder

Initializing a repository

Practise exercise:

- Open Git Bash
- Use `cd <path>` to navigate to the folder where you want to initialize the repository
- Use `git init` to initialize a new Git repository

Branching

Branches allows you to:

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The main branch (good practice):

- Core of the project
- Stable version only
- Only merge when you are sure the changes are good

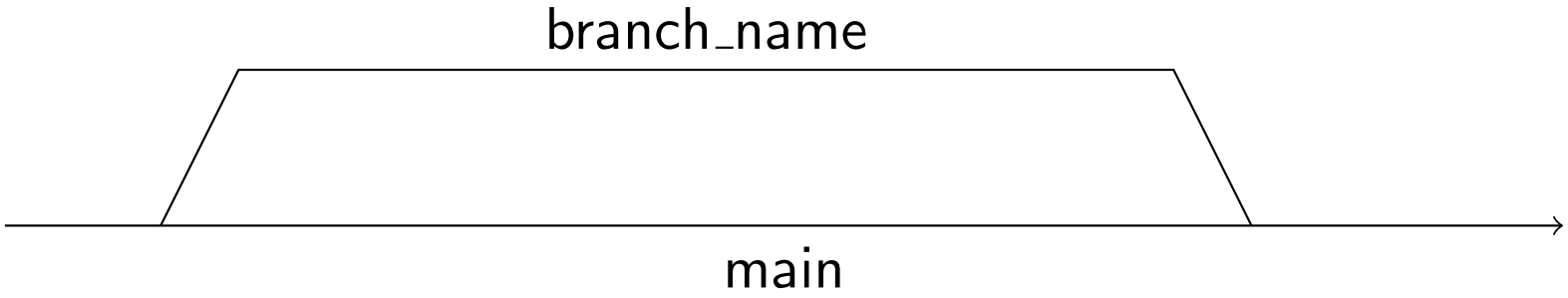
Working with branches

Practise exercise:

- Go the folder where you initialized the repository
- Create a .txt file and add some content to it
- Use `git add <file_name>` to stage the file
- Use `git commit -m "Initial commit"` to commit the changes
- Use `git branch <branch_name>` to create a new branch
- Use `git checkout <branch_name>` to switch to the new branch
- Make some changes to the .txt file and commit the changes
- Use `git checkout main` to switch back to the main branch
- Use `git merge <branch_name>` to merge the changes from the new branch to the main branch
- Use `git branch -d <branch_name>` to delete the new branch

Working with branches (cont.)

With all this, you should have a Git history that looks like this:



Be cautious, `git merge` merge the selected branch into the current branch, try to avoid this:

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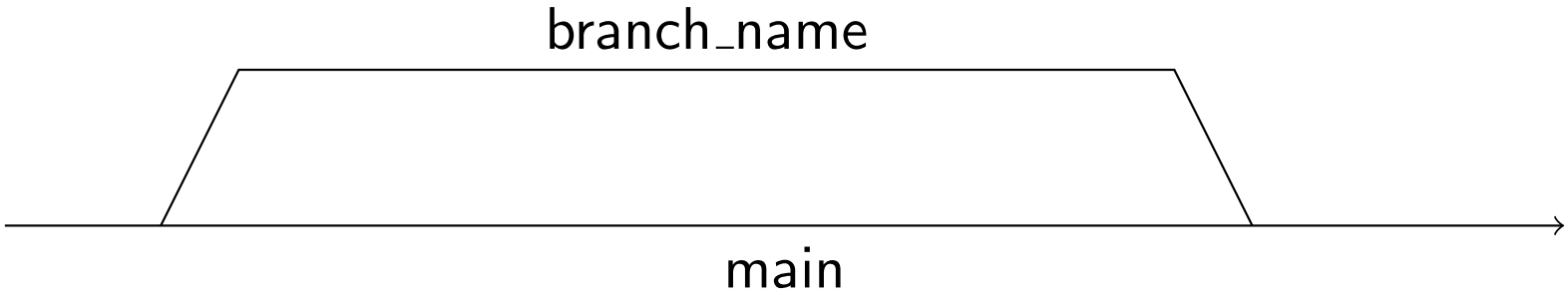

Be cautious, `git merge` merge the selected branch into the current branch, try to avoid this:

Linking local and remote
(IDE and Desktop app)

Github Desktop

If you want to work with Github and have a nice GUI:

- Dowload Github Desktop: https://desktop.github.com/
- Link your Github account
- Clone the repository you created in the previous exercises
- Make some changes to the .txt file and commit the changes
- Push the changes to the remote repository
- Go to the Github website and check that the changes are there

Git Kraken

If you want to work with GitLab/Github/Gitbucket and have a nice GUI:

- Dowload Git Kraken: https://www.gitkraken.com/
- Link your Github/GitLab/Gitbucket account
- Clone the repository you created in the previous exercises
- Make some changes to the .txt file and commit the changes
- Push the changes to the remote repository
- Go to the Github/GitLab/Gitbucket website and check that the changes are there

IDE

In most IDE, you can link your GitHub/GitLab account and work with Git directly from the IDE.

- Visual Studio Code
- JetBrains IDEs (PyCharm, IntelliJ, etc.)
- Eclipse

SSH keys and Personal Access Tokens (PAT)

When you want to push your changes to a remote repository, you need to authenticate yourself.

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- SSH keys
  - More secure than username/password
  - Can be used for multiple repositories
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- SSH keys
  - More secure than username/password
  - Can be used for multiple repositories
  - Can be used for multiple platforms (Github, GitLab, etc.)
- Personal Access Tokens (PAT)
  - Used when SSH keys are not supported (e.g., Overleaf)
  - Can be used for multiple repositories and platforms (Github, GitLab, etc.)
  - Less secure than SSH keys, but still better than username/password
  - Can be revoked at any time
  - Can have specific permissions (e.g., read-only, write-only, etc.)
  - Can be used for automation (e.g., CI/CD pipelines, scripts, etc.)
  - Can be used for third-party applications (e.g., Overleaf, etc.)

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- Scope your PAT at the strict minimum permissions you need
- If risks of compromise, revoke the PAT/SSH key and generate a new one immediately
- If you accidentally commit your PAT/SSH key to a Git repository, remove it from the history and revoke it immediately
- Add collaborators to your repository with the appropriate permissions
- If you really need to add a PAT/SSH key, add it as a secret

Conclusion

- Git is a powerful tool for version control and collaboration
- It is widely used in the software development industry
- It can be used with different platforms (Github, GitLab, etc.) and tools (IDE, desktop app, etc.)
- It is important to understand the basics of Git to work effectively with it
- Always keep a local copy of your most important works and be cautious when sharing your SSH keys or PAT

To go further

- Markdown files and documentation
- ‘.gitignore‘ files
- Seeting up a Git server locally